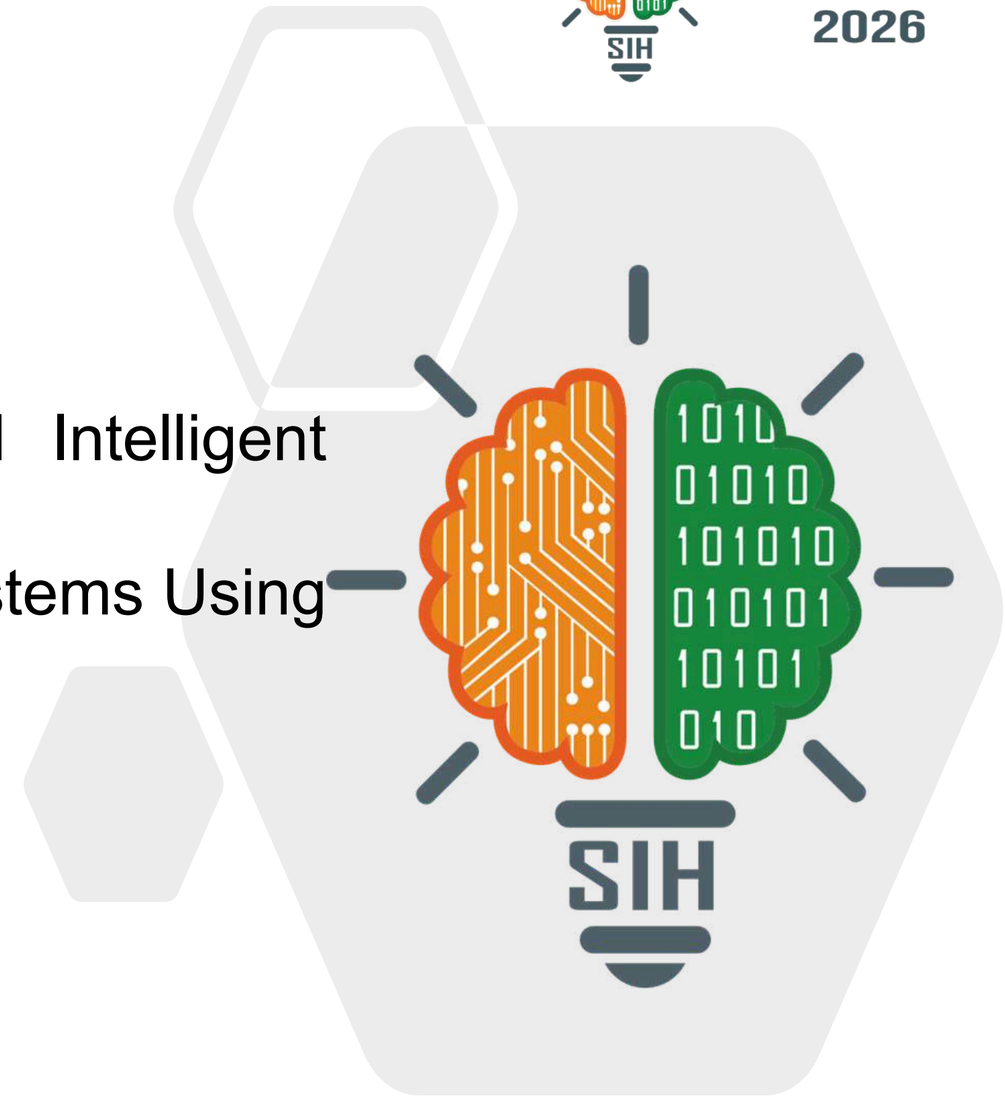
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TITLE PAGE

- **Problem Statement ID –** 261437
- **Problem Statement Title-** Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization
- **Theme-** Transportation & Logistics
- **PS Category-** Software



QRouteX

“Beyond the shortest path — Q-ROUTE continuously searches for the most efficient route as traffic conditions change.”

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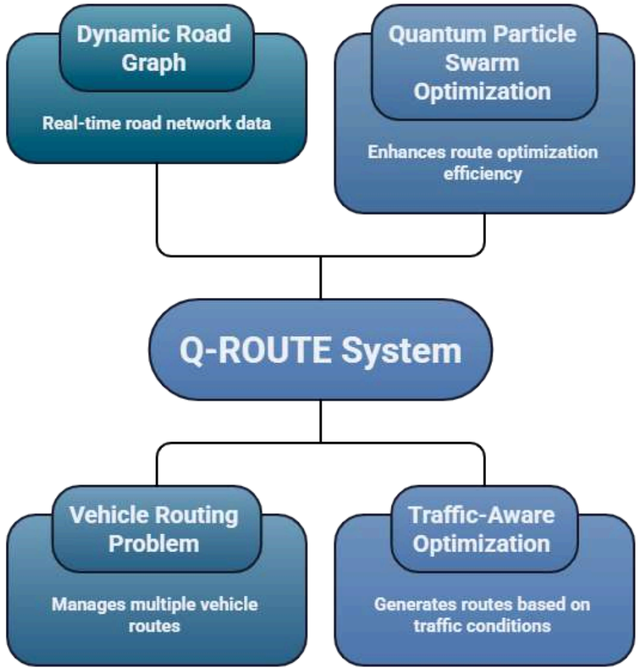
Proposed Solution & How to address the problem

🚦 **Traffic Adaptive Intelligent Routing (TAIR)**
Selects efficient routes based on current traffic and road conditions.

⚛️ **Multi-Vehicle Optimization Using QPSO**
It employs Quantum Particle Swarm Optimization to find near-optimal routes for multiple vehicles.

🚗 **Multi-Objective Route Planning**
Simultaneously optimizes the operational cost, congestion, travel time and distance

🔄 **Dynamic Re-Routing**
The routes are automatically updated and recalculated due to changes in traffic, accidents.







Q-ROUTE goes beyond finding the shortest path by using QPSO to continuously search for better multi-vehicle route combinations under changing transportation conditions.

“Traffic intelligence with QPSO optimization to find efficient routes and dynamically adapt to changing road conditions.”

QRouteX

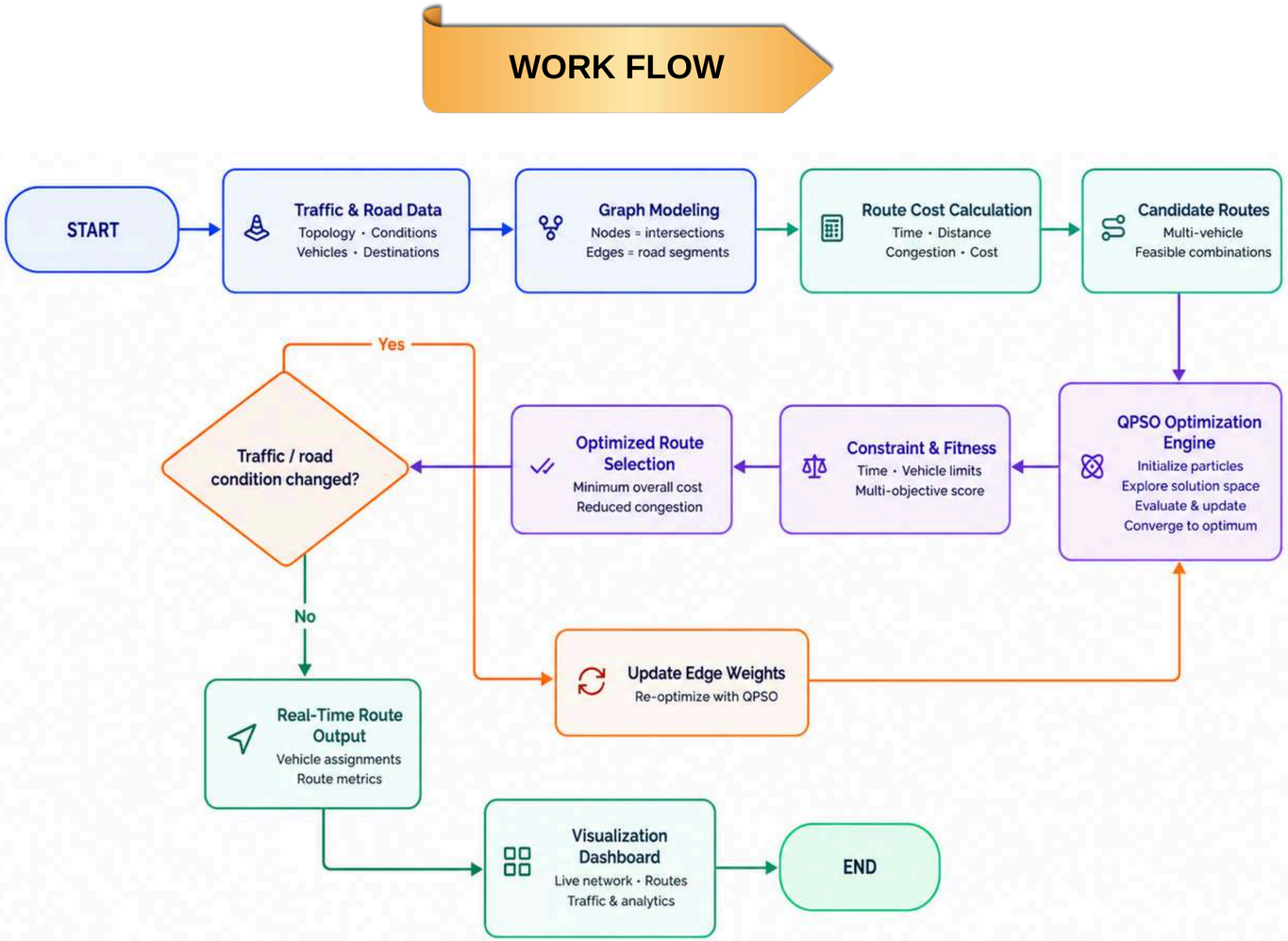
TECHNICAL APPROACH

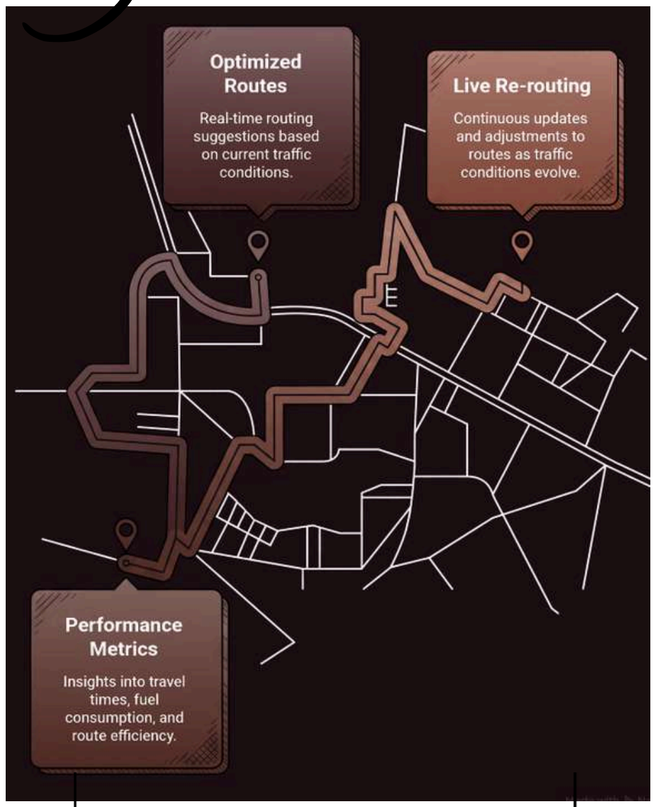
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TECH STACK

- **Python** — Core Programming
- **OSMnx** — Road Network Extraction
- **SUMO** — Traffic Simulation & Routing
- **NetworkX** — Graph Modeling & Analysis
- **QPSO** — Quantum-Inspired Optimization
- **Google OR-Tools** — Optimization & VRP Solver
- **FastAPI** — Backend API Development
- **React** — Frontend Development
- **PostgreSQL** — Database Management

WORK FLOW





"The road changes. Q-ROUTE adapts."

QRouteX

FEASIBILITY AND VIABILITY

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FASIBILITY

Software-First: Uses existing maps, traffic data & cameras with minimal hardware.

Scalable: Pilot at selected roads and expand citywide.

Low-Cost: Built using open-source tools like OSMnx, SUMO & Python.

VIABILITY

Real-World Need: Addresses congestion, delays & inefficient routing.

Benchmarkable: QIGA can be compared with Dijkstra, A* & GA.

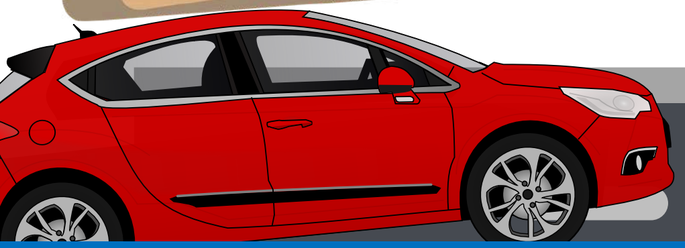
Data-Driven: Uses measurable traffic and road-condition data.

BUSINESS POTENTIAL

Cost Reduction: Saves travel time, fuel & fleet operating costs.

Smart-City Ready: Integrates with intelligent transportation systems.

Fleet Applications: Useful for logistics, buses & emergency vehicles.



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IMPACT AND BENEFITS

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IMPACTS OF Q-ROUTE



- **25% Economic Impact**
  Lower fuel costs and faster deliveries
- **25% Social Impact**
  Reduced congestion and reliable deliveries
- **25% Environmental Impact**
  Lower fuel consumption and reduced CO_2 emissions
- **25% Operational Efficiency**
  Better fleet utilization and reduced operational costs



BENEFITS OF Q-ROUTE

Reduced Travel Time
Identifies efficient routes based on changing traffic conditions.

Dynamic & Adaptive Routing
Automatically re-optimizes routes when traffic or road conditions change.



Lower Congestion & Fuel Usage
Avoids congested roads and reduces unnecessary travel.

Efficient Multi-Vehicle Planning
Optimizes routes for multiple vehicles while considering distance, cost and constraints.



QRouteX

RESEARCH AND REFERENCES

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References

Feature/Aspect	Proposed Q-ROUTE	Eclipse SUMO	Google OR-Tools
Traffic Simulation	✔	✘	✘
Real Road Network	✔	✔	✘
Multi-Objective Optimization	✔	✘	✔
Quantum-Inspired Optimization	✔	✘	✘
Dynamic Re-Routing	✔	✔	✘
Route Visualization	✔	✔	✘

DASHBOARD

